

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0588

Trumann 138 kV
June 16, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0588. The Interconnection Customer proposes a 91.5 MW generating facility interconnecting at the Trumann 138 kV substation in the Maryland area. The request was received on February 29, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,553,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Jacksonville Beach - Metzert Rd 138kV	Taber - Glenannan 345kV	312.7	176	6.3
Siuc - Jacksonville Beach 138kV	Glenannan Substation - Taber 345kV	302.9	179	7.9
Metzert Road - Alto 138kV	Taber - Glenannan 345kV	258.6	201	6.3
Alto - Laurel 138kV Line	Taber - Glenannan 345kV	202.9	229	6.3
Darien - Trumann 138kV	Sligo - Lnden 138kV	186.8	216	67.2
Laurel - Crystal Springs 138kV	Taber - Glenannan 345kV	185.7	239	6.3
Crystal Springs Substation - Taber 138kV	Taber - Glenannan Substation 345kV	159.8	254	6.3
Pangborn Substation - Derwood 138kV	Darien - Trumann 138kV	135.1	204	5.8
Pngborn Substation - Smyrna 138kV L	Darien - Trumann 138kV	119.3	215	5.8
East Orange Substation - Phillips Pump 1 138kV	Darien - Colesville 138kV	117.0	179	5.7
Flanagan - Seek Lane 138kV	Colesville - Darien 138kV	111.1	330	22.4
Story - Smyrna 138kV line	Trumnn - Darien 138kV	110.0	222	5.8
Derwood - East Orange 138kV	Colesville - Darien 138kV	107.6	179	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jacksonville Beach Substation - Metzerott Rd 138kV	Rebuild line with higher-capacity conductor	\$1,106,000	30% (\$337,000)
Siuc - Jacksonville Beach 138kV	Rebuild line with higher-capacity conductor	\$2,178,000	37% (\$801,000)
Metzerott Road - Alto 138kV	Rebuild line with higher-capacity conductor	\$12,391,000	30% (\$3,779,000)
Alto - Laurel 138kV Line	Rebuild line with higher-capacity conductor	\$3,775,000	30% (\$1,151,000)
Darien - Trumann 138kV Line	Rebuild line with higher-capacity conductor	\$871,000	100% (\$871,000)
Laurel Substation - Crystal Springs 138kV line	Rebuild line with higher-capacity conductor	\$4,817,000	30% (\$1,469,000)
Crystal Springs Substation - Taber 138kV	Rebuild line with higher-capacity conductor	\$2,453,000	30% (\$748,000)
Pangborn Substation - Derwood 138kV	Rebuild line with higher-capacity conductor	\$13,422,000	100% (\$13,422,000)
Pngborn Substation - Smyrna 138kV L	Rebuild line with higher-capacity conductor	\$9,097,000	100% (\$9,097,000)
East Orange Substation - Phillips Pump 1 138kV	Rebuild line with higher-capacity conductor	\$1,739,000	100% (\$1,739,000)
Seek Lane Substation - Flanagan 138kV	Rebuild line with higher-capacity conductor	\$1,010,000	12% (\$125,000)
Story - Smyrna 138kV line	Rebuild line with higher-capacity conductor	\$12,710,000	100% (\$12,710,000)
Derwood - East Orange 138kV	Reconductor line	\$1,304,000	100% (\$1,304,000)
	Total allocated to project		\$47,553,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.